

INERTIA

A body cannot change its state of rest or uniform motion along a straight line. This property is called inertia. Inertia has no unit and no dimension.

INERTIA

Inertia of Rest Inertia of Motion Inertia of Direction

- Inertia of Rest
Inability to change state of rest by itself.
- Inertia of Motion
Inability of a body to change its state of uniform motion by itself.
- Inertia of Direction
Inability of a body to change direction of motion by itself.

Newton's Second Law

F_{net} = Rate of change of linear momentum.

Instantaneous

$$\vec{F} = \frac{d\vec{p}}{dt}$$

Average

$$\vec{F}_{av} = \frac{\Delta \vec{p}}{\Delta t} = \frac{\vec{p}_f - \vec{p}_i}{\Delta t}$$

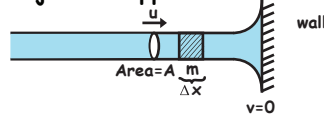
MOMENTUM

$$\vec{p} = m\vec{v}$$

- It is a vector quantity having direction same as that of velocity
- Unit is kg m/s.

LIQUID JETS

When jet is stopped at wall



$$F_{jet} = \frac{P_f - P_i}{\Delta t} = \frac{m \times 0 - mu}{\Delta t} = -\rho A u^2 \quad (m = \rho A \Delta x, \frac{\Delta x}{\Delta t} = u)$$

$$F_{wall} = \rho A u^2$$

When liquid jet bounce back

$$F_{jet} = \frac{P_f - P_i}{\Delta t} = \frac{-2mu}{\Delta t} = -2\rho A u^2$$

$$F_{wall} = 2\rho A u^2$$

When liquid jet strikes obliquely

$$F_{jet} = -2\rho A u^2 \cos \theta$$

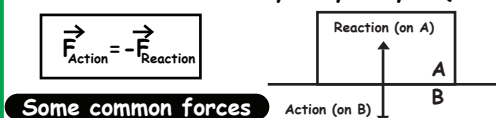
$$F_{wall} = 2\rho A u^2 \cos \theta$$

$$\text{Change in momentum} = -2mu \cos \theta$$

NEWTON'S THIRD LAW

- To every action, there is always an equal (in magnitude) and opposite (in direction) reaction.
- Forces in nature always occur in pairs.
- A single isolated force is not possible.
- Counter force experienced by a body - reaction
- Action and reaction never act on the same body

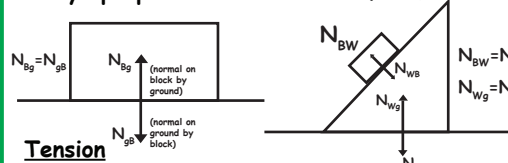
- * Force exerted on body A by body B (action)
- * force exerted on body B by body A (reaction)



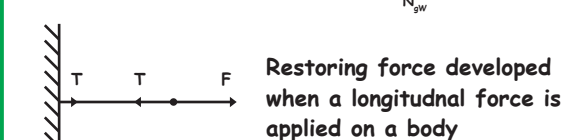
Some common forces

1. Normal Reaction

- Occurs when two surfaces are in contact with each other
- Always perpendicular to the surface.

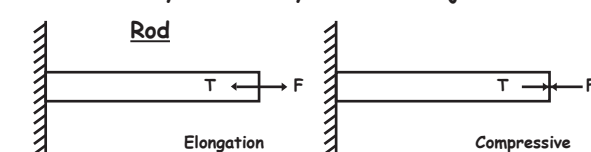


Tension



Ideal Rope

- * Massless
- * Tension is same everywhere
- * Opposes only elongation
- * On compression it becomes slack.
- * Tension always acts away from the object.



can support both elongation and compression

SINGLE BLOCK

Horizontal Force

Acceleration is along x-axis only

Along y-axis $a_y = 0$

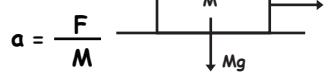
$$N - Mg = Ma_y$$

$$\therefore a_y = 0$$

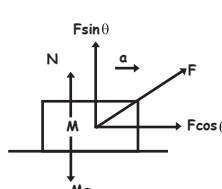
$$N = Mg$$

Along x-axis

$$F - 0 = Ma$$



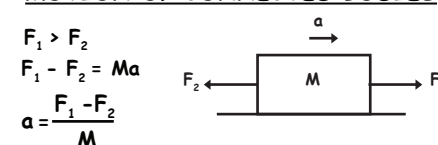
Inclined Forces



- If, $F \sin \theta < Mg$
block remains in contact with ground & accelerates horizontally
- If, $F \sin \theta = Mg$
block just leaves contact with ground

- If, $F \sin \theta > Mg$
the block leaves contact with ground and it begins to accelerate obliquely.

MOTION OF CONNECTED BODIES



Condition	Free body diagram	Equation	Force and acceleration
		$F - f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$f = m_2 a$	$f = \frac{m_2 F}{m_1 + m_2}$
		$f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - f = m_2 a$	$f = \frac{m_1 F}{m_1 + m_2}$
		$F - f_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$f_1 - f_2 = m_2 a$	$f_1 = \frac{(m_2 + m_3)F}{m_1 + m_2 + m_3}$
		$f_2 = m_3 a$	$f_2 = \frac{m_3 F}{m_1 + m_2 + m_3}$

MOTION OF BLOCKS CONNECTED BY MASSLESS STRING

Condition	Free body diagram	Equation	Tension and acceleration
		$T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - T = m_2 a$	$T = \frac{m_1 F}{m_1 + m_2}$
		$F - T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$T = m_2 a$	$T = \frac{m_2 F}{m_1 + m_2}$
		$F - T_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$T_1 - T_2 = m_2 a$	$T_1 = \frac{m_1 F}{m_1 + m_2 + m_3}$
		$F - T_2 = m_3 a$	$T_2 = \frac{(m_2 + m_3)F}{m_1 + m_2 + m_3}$

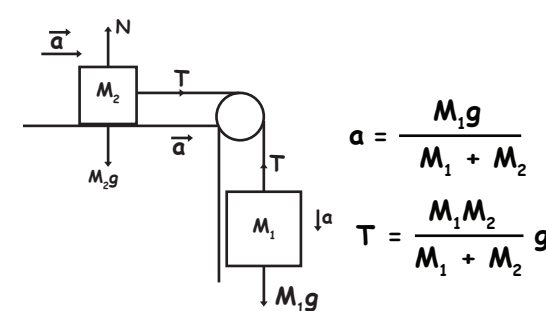
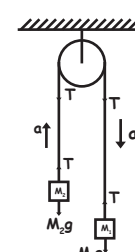
PULLEY-BLOCK SYSTEM

Ideal pulley

$$M_1 > M_2$$

$$a = \frac{F}{M} = \frac{M_1 - M_2}{M_1 + M_2} g$$

$$T = \frac{2M_1 M_2}{M_1 + M_2} g$$



INCLINED PLANE + PULLEY

		$m \mu = T - m g \sin \theta$	$a = \frac{m_2 - m_1 \sin \theta}{m_1 + m_2} g$
		$m \mu = m g \sin \theta - T$	$T = \frac{m_1 m_2 (\sin \theta + \sin \theta)}{m_1 + m_2} g$
		$m \mu \sin \theta - T = m_2 a$	$a = \frac{m_2 \sin \theta}{m_1 + m_2} g$
		$T - m_1 g \sin \theta = m_1 a$	$T = \frac{m_1 m_2 \sin \theta}{m_1 + m_2} g$

THICK ROPE

Tension will be different at different points.



$$\text{Mass per unit length} = \frac{M}{L}$$

$$\text{Mass of } x \text{ length of rope} = \frac{M}{L} x$$

Note :
Mass of given length = $\frac{\text{total mass}}{\text{total length}} \times \text{given length}$
 $\frac{\text{mass}}{\text{length}} = \text{constant}$

$$a_{rope} = \frac{F}{M}$$

$$\text{For } (L-x) \text{ rope length, } \vec{F} = m\vec{a} \Rightarrow T = m_2 \frac{F}{M}$$

LIFT PROBLEMS

Apparent weight of body in a lift

Reading of weighing machine = reaction force exerted by weighing machine

Apparent weight ($W_{apparent}$) = Reaction force (R)

Case 1: Lift is at rest

$$R = mg, W_{apparent} = W_{actual} = mg$$

Case 2: Lift moving up or down with constant velocity

$$R = mg, W_{apparent} = W_{actual} = mg$$

Case 3: Accelerated upward at a rate of 'a'

$$R - mg = ma \Rightarrow R = m(g+a) = W_{app}$$

$$W_{apparent} > W_{actual} \rightarrow \text{Feels over weight}$$

Accelerated upward at a rate of 'g'

$$R - mg = mg, R = 2mg, W_{app} = 2 \times W_{act}$$

Case 4: Accelerated downward at a rate of 'a'

$$mg - R = ma, R = m(g-a) = W_{app}, W_{app} < W_{act}$$

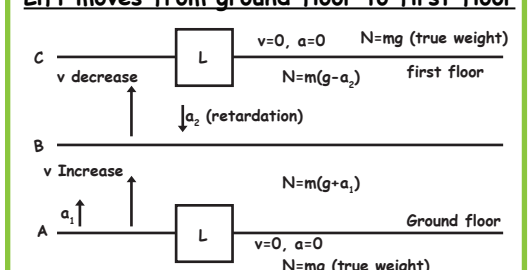
Accelerated downward at a rate of 'g' [Freefall]

$$mg - R = mg, R = mg - mg = 0, W_{app} = 0$$

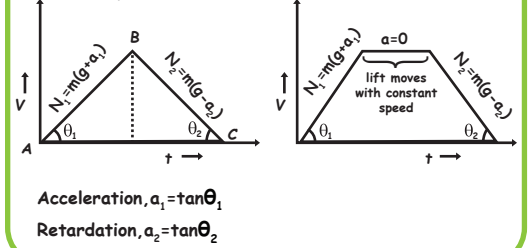
If $a > g$:

body leaves contact from ground and begins free-fall

Lift moves from ground floor to first floor



Graph



FRAME OF REFERENCE & PSEUDO FORCE

Frame of Reference

A frame in which observer is situated and makes his observation

Inertial frame of reference	Non-Inertial frame of reference
At rest or moving with uniform velocity along straight line. i.e unaccelerated	Accelerated frame of reference.
Newton's law of motion hold's $\vec{F}_{net} = m\vec{a}$	Newton's law of motion not applicable. $\vec{F}_{net} + \vec{F}_{pseudo} = m\vec{a}$ $\vec{F}_{pseudo} = -m\vec{a}_0$

LAWS OF MOTION

RELATIVE SLIPPING

Minimum force required to push the inclined plane such that "m" does not slip with respect to "M"

$$F = (m+M) g \tan \theta, \quad a = g \tan \theta$$

Minimum acceleration of "M" such that there is no relative slipping.

$$a = \left(\frac{m_1}{m_2}\right) g$$

Minimum mass M_0 such that there is no relative slipping

$$M_0 = \frac{M+m}{\cot \theta - 1}$$

EQUILIBRIUM & LAMI'S THEOREM

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

PARALLELOGRAM LAW

$$\vec{F} = \vec{F}_1 + \vec{F}_2, \quad F = \sqrt{F_1^2 + F_2^2}, \quad \tan \theta = \frac{F_2}{F_1}$$

MAN-CAGE PROBLEM

Man holds the cage stationary

$$T = \left(\frac{m+M}{2}\right) g$$

$$N = \left(\frac{m-M}{2}\right) g$$

- $m < M$
- if $m > M$ man cannot hold the cage stationary

IMPULSE

If a large force acting for short period of time, there will be a sudden change in momentum

$$\vec{F}_{imp} = \frac{d\vec{p}}{dt}$$

$$\vec{p}_f - \vec{p}_i = \int_0^t \vec{F}_{imp} dt = \text{area under } F-t \text{ graph}$$

$$\vec{I} = \text{Impulse} = \vec{p}_f - \vec{p}_i = \int_0^t \vec{F}_{imp} dt = \text{area of } F-t \text{ graph}$$

Case-1

Impulse

$$I = p_f - p_i = mV_2 - (-mV_1)$$

$$I = m(V_1 + V_2)$$

Average impulsive force

$$F_{avg} = \frac{\text{impulse}}{t_o} = \frac{\Delta p}{t_o} = \frac{m(V_1 + V_2)}{t_o}$$

Case-2

Impulse

$$m(v+v) = J$$

$$J = m(2v)$$

Average impulsive force

$$F_{avg} = \frac{\Delta p}{\Delta t} = \frac{2mv}{t_o}$$

ROD SLIDING ON A WALL

velocity of B towards A = $v \sin \theta$

velocity of A away from B = $u \cos \theta$

these velocities should be equal,

$$v \sin \theta = u \cos \theta$$

$$v = u \cot \theta$$

FRICTION

Static friction

- It is a self adjusting force.
- The opposing force that comes into play, when object tends to slip over the surface of other object, but slipping has not yet started.
- As applied force increases static friction also increases.
- The body doesn't move until a maximum value of static friction is attained
- This value is called limiting friction or $(f_s)_{max}$

$$(f_s)_{max} = \mu_s N$$

Kinetic friction

If the applied force is increased further and slipping between surfaces start, the friction opposing the slipping is called kinetic friction.

Static region

Kinetic region

PULLING FORCE & PUSHING FORCE

Pulling force

$$F \cos \theta \geq \mu_s (mg - F \sin \theta)$$

$$F \geq \frac{\mu_s mg}{\cos \theta + \mu_s \sin \theta}$$

$$F_{pulling} = \frac{\mu_s mg}{\cos \theta + \mu_s \sin \theta}$$

Pushing force

$$F \cos \theta \geq \mu_s (mg + F \sin \theta)$$

$$F \geq \frac{\mu_s mg}{\cos \theta - \mu_s \sin \theta}$$

$$F_{pushing} = \frac{\mu_s mg}{\cos \theta - \mu_s \sin \theta}$$

$F_{pushing} > F_{pulling}$

ANGLE OF FRICTION

Angle(θ) made by resultant of normal (N) & frictional force(f_s) with normal

$$R = \sqrt{N^2 + f_s^2}$$

R is resultant

$$\tan \theta = \frac{f_s}{N}$$

When $f_s = (f_s)_{max}$ $R = N \sqrt{1 + \mu_s^2}$

$$\tan \theta = \mu_s, \quad \mu_s \text{ is Coefficient of friction}$$

i) $f = 0$

$$\tan \theta = 0$$

$$\theta = 0$$

ii) $0 < f_s < (f_s)_{max}$

$$0 \leq \tan \theta \leq \mu_s$$

$$N < R < N \sqrt{1 + \mu_s^2}$$

iii) $f_s = (f_s)_{max}$

$$R_{max} = N \sqrt{1 + \mu_s^2}$$

$$\tan \theta_{max} = \mu_s$$

iv) When slipping has started,

$$f = f_k$$

$$R_{max} = N \sqrt{1 + \mu_k^2}$$

θ = Angle of kinetic friction

ANGLE OF REPOSE

Angle made by inclined plane such that a block kept on it just begins to slide

Depends only on μ_s and is independent of mass.

$$\tan \theta_0 = \mu_s$$

Variation of R

As angle of inclined plane increases, R remains constant (=mg) and when sliding starts R starts decreasing.

Variation of angle of friction.

As angle of inclined plane increases, angle of friction will also increase and as sliding starts its value becomes constant and $\tan \theta = \mu_k$

Minimum & Maximum force (applied parallel to inclined plane)

$$F_{min} = mg (\sin \theta - \mu_s \cos \theta)$$

F_{min} is the minimum force required to keep the block stationary if the angle of inclined plane is greater than angle of repose

$$F_{max} = mg (\sin \theta + \mu_s \cos \theta)$$

$mg (\sin \theta - \mu_s \cos \theta) \leq F \leq mg (\sin \theta + \mu_s \cos \theta)$

HORIZONTAL TRUCK BOX

Case-1

Box does not slip.

$$f_s \leq \mu_s N$$

$$\Rightarrow ma_T \leq \mu_s mg$$

$$\Rightarrow a_T \leq \mu_s g$$

Case-2

Box slips

$$a_T \geq \mu_k g$$

VERTICAL TRUCK BOX

Case-1

Box does not slip.

$$f_s \leq \mu_s N$$

$$\Rightarrow mg \leq \mu_s ma_T$$

$$\Rightarrow \mu_s \geq \frac{g}{a_T}$$

Case-2

Box slips

$$\mu_k \leq \frac{g}{a_T}$$

LAWS OF MOTION

